
HD44780U (LCD-II)

(Dot Matrix Liquid Crystal Display Controller/Driver)

HITACHI

ADE-207-272(Z)

'99.9

Rev. 0.0

Description

The HD44780U dot-matrix liquid crystal display controller and driver LSI displays alphanumerics, Japanese kana characters, and symbols. It can be configured to drive a dot-matrix liquid crystal display under the control of a 4- or 8-bit microprocessor. Since all the functions such as display RAM, character generator, and liquid crystal driver, required for driving a dot-matrix liquid crystal display are internally provided on one chip, a minimal system can be interfaced with this controller/driver.

A single HD44780U can display up to one 8-character line or two 8-character lines.

The HD44780U has pin function compatibility with the HD44780S which allows the user to easily replace an LCD-II with an HD44780U. The HD44780U character generator ROM is extended to generate 208×8 dot character fonts and $32 \times 5 \times 10$ dot character fonts for a total of 240 different character fonts.

The low power supply (2.7V to 5.5V) of the HD44780U is suitable for any portable battery-driven product requiring low power dissipation.

Features

- 5×8 and 5×10 dot matrix possible
- Low power operation support:
 - 2.7 to 5.5V
- Wide range of liquid crystal display driver power
 - 3.0 to 11V
- Liquid crystal drive waveform
 - A (One line frequency AC waveform)
- Correspond to high speed MPU bus interface
 - 2 MHz (when $V_{CC} = 5V$)
- 4-bit or 8-bit MPU interface enabled
- 80×8 -bit display RAM (80 characters max.)
- 9,920-bit character generator ROM for a total of 240 character fonts
 - 208 character fonts (5×8 dot)
 - 32 character fonts (5×10 dot)

HD44780U

- 64 × 8-bit character generator RAM
 - 8 character fonts (5 × 8 dot)
 - 4 character fonts (5 × 10 dot)
- 16-common × 40-segment liquid crystal display driver
- Programmable duty cycles
 - 1/8 for one line of 5 × 8 dots with cursor
 - 1/11 for one line of 5 × 10 dots with cursor
 - 1/16 for two lines of 5 × 8 dots with cursor
- Wide range of instruction functions:
 - Display clear, cursor home, display on/off, cursor on/off, display character blink, cursor shift, display shift
- Pin function compatibility with HD44780S
- Automatic reset circuit that initializes the controller/driver after power on
- Internal oscillator with external resistors
- Low power consumption

Ordering Information

Type No.	Package	CGROM
HD44780UA00FS	FP-80B	Japanese standard font
HCD44780UA00	Chip	
HD44780UA00TF	TFP-80F	
HD44780UA02FS	FP-80B	European standard font
HCD44780UA02	Chip	
HD44780UA02TF	TFP-80F	
HD44780UBxxFS	FP-80B	Custom font
HCD44780UBxx	Chip	
HD44780UBxxTF	TFP-80F	

Note: xx: ROM code No.

Function Description

Registers

The HD44780U has two 8-bit registers, an instruction register (IR) and a data register (DR).

The IR stores instruction codes, such as display clear and cursor shift, and address information for display data RAM (DDRAM) and character generator RAM (CGRAM). The IR can only be written from the MPU.

The DR temporarily stores data to be written into DDRAM or CGRAM and temporarily stores data to be read from DDRAM or CGRAM. Data written into the DR from the MPU is automatically written into DDRAM or CGRAM by an internal operation. The DR is also used for data storage when reading data from DDRAM or CGRAM. When address information is written into the IR, data is read and then stored into the DR from DDRAM or CGRAM by an internal operation. Data transfer between the MPU is then completed when the MPU reads the DR. After the read, data in DDRAM or CGRAM at the next address is sent to the DR for the next read from the MPU. By the register selector (RS) signal, these two registers can be selected (Table 1).

Busy Flag (BF)

When the busy flag is 1, the HD44780U is in the internal operation mode, and the next instruction will not be accepted. When $RS = 0$ and $R/\overline{W} = 1$ (Table 1), the busy flag is output to DB7. The next instruction must be written after ensuring that the busy flag is 0.

Address Counter (AC)

The address counter (AC) assigns addresses to both DDRAM and CGRAM. When an address of an instruction is written into the IR, the address information is sent from the IR to the AC. Selection of either DDRAM or CGRAM is also determined concurrently by the instruction.

After writing into (reading from) DDRAM or CGRAM, the AC is automatically incremented by 1 (decremented by 1). The AC contents are then output to DB0 to DB6 when $RS = 0$ and $R/\overline{W} = 1$ (Table 1).

Table 1 Register Selection

RS	R/ $\overline{W}$	Operation
0	0	IR write as an internal operation (display clear, etc.)
0	1	Read busy flag (DB7) and address counter (DB0 to DB6)
1	0	DR write as an internal operation (DR to DDRAM or CGRAM)
1	1	DR read as an internal operation (DDRAM or CGRAM to DR)

- Case 2: For a 16-character × 2-line display, the HD44780 can be extended using one 40-output extension driver. See Figure 6.
- When display shift operation is performed, the DDRAM address shifts. See Figure 6.

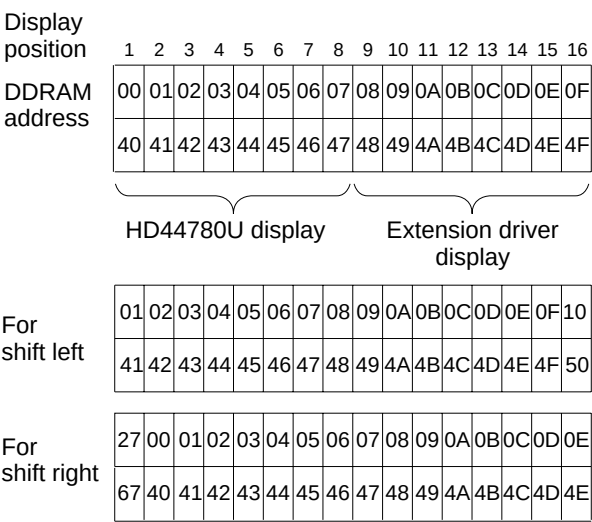


Figure 6 2-Line by 16-Character Display Example

Table 4 Correspondence between Character Codes and Character Patterns (ROM Code: A00)

Lower 4 Bits	Upper 4 Bits		0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111	
	CG RAM (1)					0	@	P	`	P				一	夕	三	α	p	
xxxx0000																			
xxxx0001	(2)				!	1	A	Q	a	4				a	7	+	4	ä	q
xxxx0010	(3)				"	2	B	R	b	r				「	イ	ウ	×	β	θ
xxxx0011	(4)				#	3	C	S	c	s				」	ウ	テ	ε	ω	
xxxx0100	(5)				\$	4	D	T	d	t				、	エ	ト	ト	μ	Ω
xxxx0101	(6)				%	5	E	U	e	u				・	オ	ナ	1	ε	Ü
xxxx0110	(7)				&	6	F	V	f	v				ヲ	カ	ニ	ヨ	ρ	Σ
xxxx0111	(8)				'	7	G	W	g	w				ア	キ	ヌ	ウ	g	π
xxxx1000	(1)				(	8	H	X	h	x				イ	ク	ネ	リ	フ	Σ
xxxx1001	(2)				)	9	I	Y	i	y				ウ	ク	ル	ル	´	Y
xxxx1010	(3)				*	:	J	Z	j	z				エ	コ	ン	レ	j	≠
xxxx1011	(4)				+	:	K	C	k	c				オ	サ	ヒ	ロ	*	π
xxxx1100	(5)				,	<	L	*	1	1				ト	シ	フ	フ	Φ	π
xxxx1101	(6)				—	=	M	I	m	}				ユ	ズ	ハ	ン	±	÷
xxxx1110	(7)				.	>	N	^	n	÷				ヨ	セ	ホ	°	ñ	
xxxx1111	(8)				/	?	O	_	o	+				ッ	リ	マ	°	ö	

Note: The user can specify any pattern for character-generator RAM.

Table 4 Correspondence between Character Codes and Character Patterns (ROM Code: A02)

Lower 4 Bits \ Upper 4 Bits		0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111
xxxx0000		CG RAM (1)	⬮		0	Q	P	`	P	E	α	≡	°	À	Ⓐ	À	⌘
xxxx0001	(2)		⬮	!	1	A	Q	a	9	A	Ⓐ	i	±	Á	Ñ	á	ñ
xxxx0010	(3)		“	"	2	B	R	b	r	⌘	Γ	¢	²	Â	Ò	â	ò
xxxx0011	(4)		”	#	3	C	S	c	s	3	π	£	³	Ã	Ó	ã	ó
xxxx0100	(5)		⬮	\$	4	D	T	d	t	⌘	Σ	×	ℝ	Ä	Ô	ä	ô
xxxx0101	(6)		⬮	%	5	E	U	e	u	Ÿ	σ	¥	ℳ	Å	Ö	å	ö
xxxx0110	(7)		⬮	&	6	F	V	f	v	⌘	Ⓐ	!	9	Æ	Ö	æ	ö
xxxx0111	(8)		⬮	'	7	G	W	g	w	⌘	τ	§	•	Ç	×	ç	÷
xxxx1000	(1)		⬮	(	8	H	X	h	x	Ÿ	⌘	†	ω	È	⌘	è	⌘
xxxx1001	(2)		⬮	)	9	I	Y	i	y	⌘	Ⓐ	¹	¹	É	Ù	é	ù
xxxx1010	(3)		⬮	*	:	J	Z	j	z	4	Ω	Ⓐ	Ⓐ	Ê	Ú	ê	ú
xxxx1011	(4)		⬮	+	;	K	⌘	k	(	⌘	δ	⌘	⌘	Ë	Û	ë	û
xxxx1100	(5)		⬮	,	<	L	\	l	l	⌘	∞	⌘	⌘	Ï	Ü	ï	ü
xxxx1101	(6)		⬮	-	=	M	⌘	m	>	⌘	⌘	⌘	⌘	Í	Ý	í	ý
xxxx1110	(7)		⬮	.	>	N	^	n	~	⌘	ε	⌘	⌘	Î	Ë	î	ë
xxxx1111	(8)		⬮	/	?	O	_	o	ó	⌘	⌘	‘	¿	İ	ß	ı	ı

Instruction Description

Clear Display

Clear display writes space code 20H (character pattern for character code 20H must be a blank pattern) into all DDRAM addresses. It then sets DDRAM address 0 into the address counter, and returns the display to its original status if it was shifted. In other words, the display disappears and the cursor or blinking goes to the left edge of the display (in the first line if 2 lines are displayed). It also sets I/D to 1 (increment mode) in entry mode. S of entry mode does not change.

Return Home

Return home sets DDRAM address 0 into the address counter, and returns the display to its original status if it was shifted. The DDRAM contents do not change.

The cursor or blinking go to the left edge of the display (in the first line if 2 lines are displayed).

Entry Mode Set

I/D: Increments ($I/D = 1$) or decrements ($I/D = 0$) the DDRAM address by 1 when a character code is written into or read from DDRAM.

The cursor or blinking moves to the right when incremented by 1 and to the left when decremented by 1. The same applies to writing and reading of CGRAM.

S: Shifts the entire display either to the right ($I/D = 0$) or to the left ($I/D = 1$) when S is 1. The display does not shift if S is 0.

If S is 1, it will seem as if the cursor does not move but the display does. The display does not shift when reading from DDRAM. Also, writing into or reading out from CGRAM does not shift the display.

Display On/Off Control

D: The display is on when D is 1 and off when D is 0. When off, the display data remains in DDRAM, but can be displayed instantly by setting D to 1.

C: The cursor is displayed when C is 1 and not displayed when C is 0. Even if the cursor disappears, the function of I/D or other specifications will not change during display data write. The cursor is displayed using 5 dots in the 8th line for 5×8 dot character font selection and in the 11th line for the 5×10 dot character font selection (Figure 13).

B: The character indicated by the cursor blinks when B is 1 (Figure 13). The blinking is displayed as switching between all blank dots and displayed characters at a speed of 409.6-ms intervals when f_{cp} or f_{osc} is 250 kHz. The cursor and blinking can be set to display simultaneously. (The blinking frequency changes according to f_{osc} or the reciprocal of f_{cp} . For example, when f_{cp} is 270 kHz, $409.6 \times 250/270 = 379.2$ ms.)

Cursor or Display Shift

Cursor or display shift shifts the cursor position or display to the right or left without writing or reading display data (Table 7). This function is used to correct or search the display. In a 2-line display, the cursor moves to the second line when it passes the 40th digit of the first line. Note that the first and second line displays will shift at the same time.

When the displayed data is shifted repeatedly each line moves only horizontally. The second line display does not shift into the first line position.

The address counter (AC) contents will not change if the only action performed is a display shift.

Function Set

DL: Sets the interface data length. Data is sent or received in 8-bit lengths (DB7 to DB0) when DL is 1, and in 4-bit lengths (DB7 to DB4) when DL is 0. When 4-bit length is selected, data must be sent or received twice.

N: Sets the number of display lines.

F: Sets the character font.

Note: Perform the function at the head of the program before executing any instructions (except for the read busy flag and address instruction). From this point, the function set instruction cannot be executed unless the interface data length is changed.

Set CGRAM Address

Set CGRAM address sets the CGRAM address binary AAAAAA into the address counter.

Data is then written to or read from the MPU for CGRAM.

		RS	R/W	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0	
Clear display	Code	0	0	0	0	0	0	0	0	0	1	
Return home	Code	0	0	0	0	0	0	0	0	1	*	Note: * Don't care.
Entry mode set	Code	0	0	0	0	0	0	0	1	I/D	S	
Display on/off control	Code	0	0	0	0	0	0	1	D	C	B	
Cursor or display shift	Code	0	0	0	0	0	1	S/C	R/L	*	*	Note: * Don't care.
Function set	Code	0	0	0	0	1	DL	N	F	*	*	
Set CGRAM address	Code	0	0	0	1	A	A	A	A	A	A	

Figure 11 Instruction (1)

Set DDRAM Address

Set DDRAM address sets the DDRAM address binary AAAAAAA into the address counter.

Data is then written to or read from the MPU for DDRAM.

However, when N is 0 (1-line display), AAAAAAA can be 00H to 4FH. When N is 1 (2-line display), AAAAAAA can be 00H to 27H for the first line, and 40H to 67H for the second line.

Read Busy Flag and Address

Read busy flag and address reads the busy flag (BF) indicating that the system is now internally operating on a previously received instruction. If BF is 1, the internal operation is in progress. The next instruction will not be accepted until BF is reset to 0. Check the BF status before the next write operation. At the same time, the value of the address counter in binary AAAAAAA is read out. This address counter is used by both CG and DDRAM addresses, and its value is determined by the previous instruction. The address contents are the same as for instructions set CGRAM address and set DDRAM address.

Table 7 Shift Function

S/C	R/L	
0	0	Shifts the cursor position to the left. (AC is decremented by one.)
0	1	Shifts the cursor position to the right. (AC is incremented by one.)
1	0	Shifts the entire display to the left. The cursor follows the display shift.
1	1	Shifts the entire display to the right. The cursor follows the display shift.

Table 8 Function Set

N	F	No. of Display Lines	Character Font	Duty Factor	Remarks
0	0	1	5 × 8 dots	1/8	
0	1	1	5 × 10 dots	1/11	
1	*	2	5 × 8 dots	1/16	Cannot display two lines for 5 × 10 dot character font

Note: * Indicates don't care.

Instruction and Display Correspondence

- 8-bit operation, 8-digit $\times$ 1-line display with internal reset

Refer to Table 11 for an example of an 8-digit $\times$ 1-line display in 8-bit operation. The HD44780U functions must be set by the function set instruction prior to the display. Since the display data RAM can store data for 80 characters, as explained before, the RAM can be used for displays such as for advertising when combined with the display shift operation.

Since the display shift operation changes only the display position with DDRAM contents unchanged, the first display data entered into DDRAM can be output when the return home operation is performed.

- 4-bit operation, 8-digit $\times$ 1-line display with internal reset

The program must set all functions prior to the 4-bit operation (Table 12). When the power is turned on, 8-bit operation is automatically selected and the first write is performed as an 8-bit operation. Since DB0 to DB3 are not connected, a rewrite is then required. However, since one operation is completed in two accesses for 4-bit operation, a rewrite is needed to set the functions (see Table 12). Thus, DB4 to DB7 of the function set instruction is written twice.

- 8-bit operation, 8-digit $\times$ 2-line display

For a 2-line display, the cursor automatically moves from the first to the second line after the 40th digit of the first line has been written. Thus, if there are only 8 characters in the first line, the DDRAM address must be again set after the 8th character is completed. (See Table 13.) Note that the display shift operation is performed for the first and second lines. In the example of Table 13, the display shift is performed when the cursor is on the second line. However, if the shift operation is performed when the cursor is on the first line, both the first and second lines move together. If the shift is repeated, the display of the second line will not move to the first line. The same display will only shift within its own line for the number of times the shift is repeated.

Note: When using the internal reset, the electrical characteristics in the Power Supply Conditions Using Internal Reset Circuit table must be satisfied. If not, the HD44780U must be initialized by instructions. See the section, Initializing by Instruction.

Table 11 8-Bit Operation, 8-Digit × 1-Line Display Example with Internal Reset

Step	Instruction										Display	Operation
No.	RS	R/W	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0		
1	Power supply on (the HD44780U is initialized by the internal reset circuit)											Initialized. No display.
2	Function set 00001100**											Sets to 8-bit operation and selects 1-line display and 5 × 8 dot character font. (Number of display lines and character fonts cannot be changed after step #2.)
3	Display on/off control 0000001110										—	Turns on display and cursor. Entire display is in space mode because of initialization.
4	Entry mode set 000000110										—	Sets mode to increment the address by one and to shift the cursor to the right at the time of write to the DD/CGRAM. Display is not shifted.
5	Write data to CGRAM/DDRAM 1001001000										H—	Writes H. DDRAM has already been selected by initialization when the power was turned on. The cursor is incremented by one and shifted to the right.
6	Write data to CGRAM/DDRAM 1001001001										HI—	Writes I.
7												
8	Write data to CGRAM/DDRAM 1001001001										HITACHI—	Writes I.
9	Entry mode set 000000111										HITACHI—	Sets mode to shift display at the time of write.
10	Write data to CGRAM/DDRAM 1000100000										ITACHI _	Writes a space.

Table 11 8-Bit Operation, 8-Digit × 1-Line Display Example with Internal Reset (cont)

Step	Instruction											Display	Operation
No.	RS	R/W	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0			
11	Write data to CGRAM/DDRAM											TACHI M_	Writes M.
	1	0	0	1	0	0	1	1	0	1			
12													Writes M.
13	Write data to CGRAM/DDRAM											MICROKO_	Writes O.
	1	0	0	1	0	0	1	1	1	1			
14	Cursor or display shift											MICROK<u>O</u>	Shifts only the cursor position to the left.
	0	0	0	0	0	1	0	0	*	*			
15	Cursor or display shift											MICROK<u>O</u>	Shifts only the cursor position to the left.
	0	0	0	0	0	1	0	0	*	*			
16	Write data to CGRAM/DDRAM											ICROK<u>O</u>	Writes C over K. The display moves to the left.
	1	0	0	1	0	0	0	0	1	1			
17	Cursor or display shift											MICROK<u>O</u>	Shifts the display and cursor position to the right.
	0	0	0	0	0	1	1	1	*	*			
18	Cursor or display shift											MICROK<u>O</u>_	Shifts the display and cursor position to the right.
	0	0	0	0	0	1	0	1	*	*			
19	Write data to CGRAM/DDRAM											ICROCOM_	Writes M.
	1	0	0	1	0	0	1	1	0	1			
20													Writes M.
21	Return home											HITACHI	Returns both display and cursor to the original position (address 0).
	0	0	0	0	0	0	0	0	1	0			

Table 12 4-Bit Operation, 8-Digit × 1-Line Display Example with Internal Reset

Step		Instruction					Display	Operation
No.	RS	R/W	DB7	DB6	DB5	DB4		
1	Power supply on (the HD44780U is initialized by the internal reset circuit)							Initialized. No display.
2	Function set 0 0 0 0 1 0							Sets to 4-bit operation. In this case, operation is handled as 8 bits by initialization, and only this instruction completes with one write.
3	Function set 0 0 0 0 1 0 0 0 0 0 * *							Sets 4-bit operation and selects 1-line display and 5 × 8 dot character font. 4-bit operation starts from this step and resetting is necessary. (Number of display lines and character fonts cannot be changed after step #3.)
4	Display on/off control 0 0 0 0 0 0 0 0 1 1 1 0						—	Turns on display and cursor. Entire display is in space mode because of initialization.
5	Entry mode set 0 0 0 0 0 0 0 0 0 1 1 0						—	Sets mode to increment the address by one and to shift the cursor to the right at the time of write to the DD/CGRAM. Display is not shifted.
6	Write data to CGRAM/DDRAM 1 0 0 1 0 0 1 0 1 0 0 0						H—	Writes H. The cursor is incremented by one and shifts to the right.

Note: The control is the same as for 8-bit operation beyond step #6.

Table 13 8-Bit Operation, 8-Digit × 2-Line Display Example with Internal Reset

Step		Instruction										Display	Operation
No.	RS	R/W	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0			
1	Power supply on (the HD44780U is initialized by the internal reset circuit)											Initialized. No display.	
2	Function set 0 0 0 0 1 1 1 0 * *											Sets to 8-bit operation and selects 2-line display and 5 × 8 dot character font.	
3	Display on/off control 0 0 0 0 0 0 1 1 1 0										—	Turns on display and cursor. All display is in space mode because of initialization.	
4	Entry mode set 0 0 0 0 0 0 0 1 1 0										—	Sets mode to increment the address by one and to shift the cursor to the right at the time of write to the DD/CGRAM. Display is not shifted.	
5	Write data to CGRAM/DDRAM 1 0 0 1 0 0 1 0 0 0										H—	Writes H. DDRAM has already been selected by initialization when the power was turned on. The cursor is incremented by one and shifted to the right.	
6													
7	Write data to CGRAM/DDRAM 1 0 0 1 0 0 1 0 0 1										HITACHI—	Writes I.	
8	Set DDRAM address 0 0 1 1 0 0 0 0 0 0										HITACHI —	Sets DDRAM address so that the cursor is positioned at the head of the second line.	

Table 13 8-Bit Operation, 8-Digit × 2-Line Display Example with Internal Reset (cont)

Step	Instruction											Display	Operation
No.	RS	R/W	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0			
9	Write data to CGRAM/DDRAM											HITACHI M_	Writes M.
	1	0	0	1	0	0	1	1	0	1			
10													
11	Write data to CGRAM/DDRAM											HITACHI MICROCO_	Writes O.
	1	0	0	1	0	0	1	1	1	1			
12	Entry mode set											HITACHI MICROCO_	Sets mode to shift display at the time of write.
	0	0	0	0	0	0	0	1	1	1			
13	Write data to CGRAM/DDRAM											ITACHI ICROCOM_	Writes M. Display is shifted to the left. The first and second lines both shift at the same time.
	1	0	0	1	0	0	1	1	0	1			
14													
15	Return home											HITACHI MICROCOM	Returns both display and cursor to the original position (address 0).
	0	0	0	0	0	0	0	0	1	0			

Initializing by Instruction

If the power supply conditions for correctly operating the internal reset circuit are not met, initialization by instructions becomes necessary.

Refer to Figures 23 and 24 for the procedures on 8-bit and 4-bit initializations, respectively.

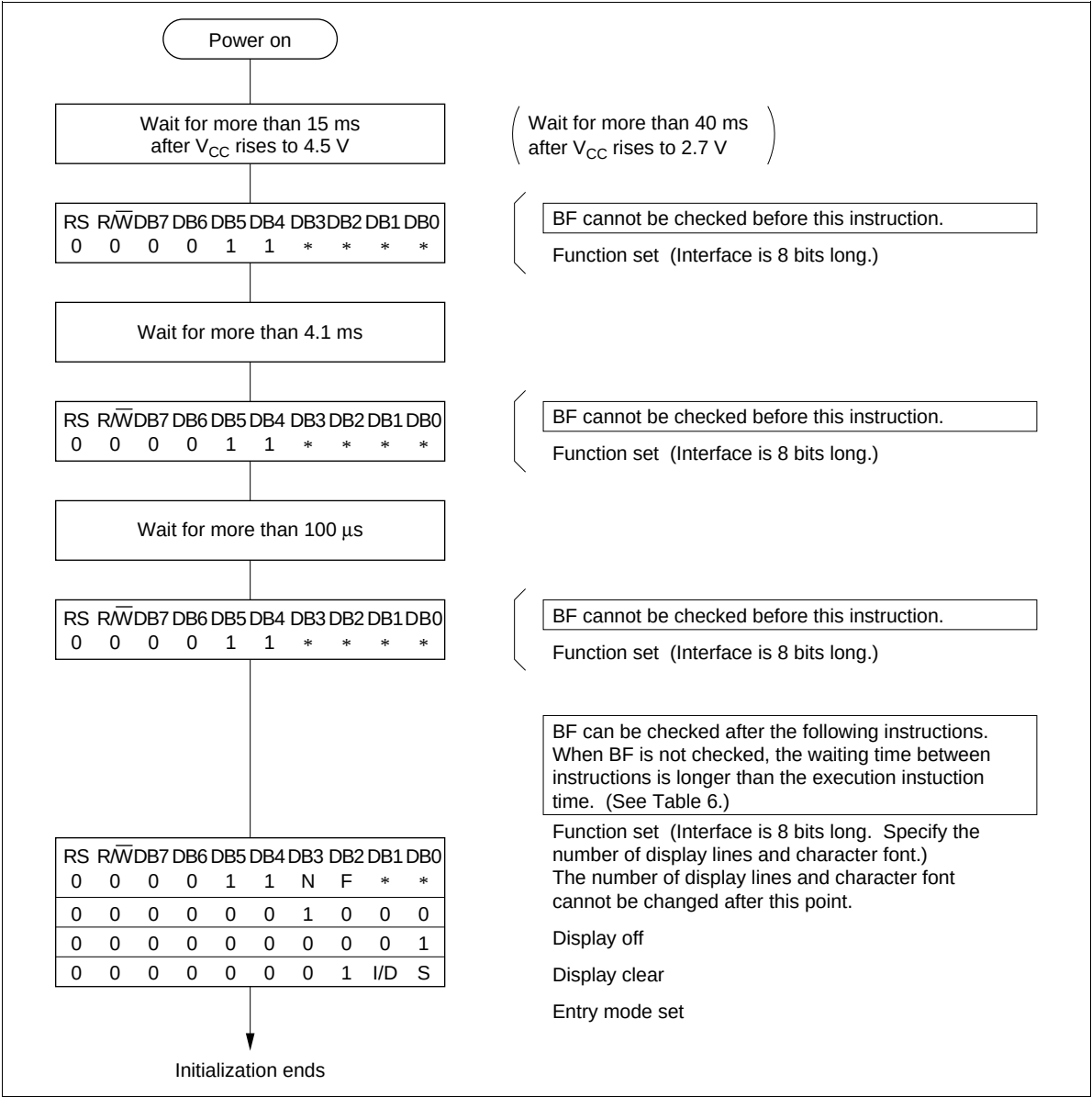


Figure 23 8-Bit Interface

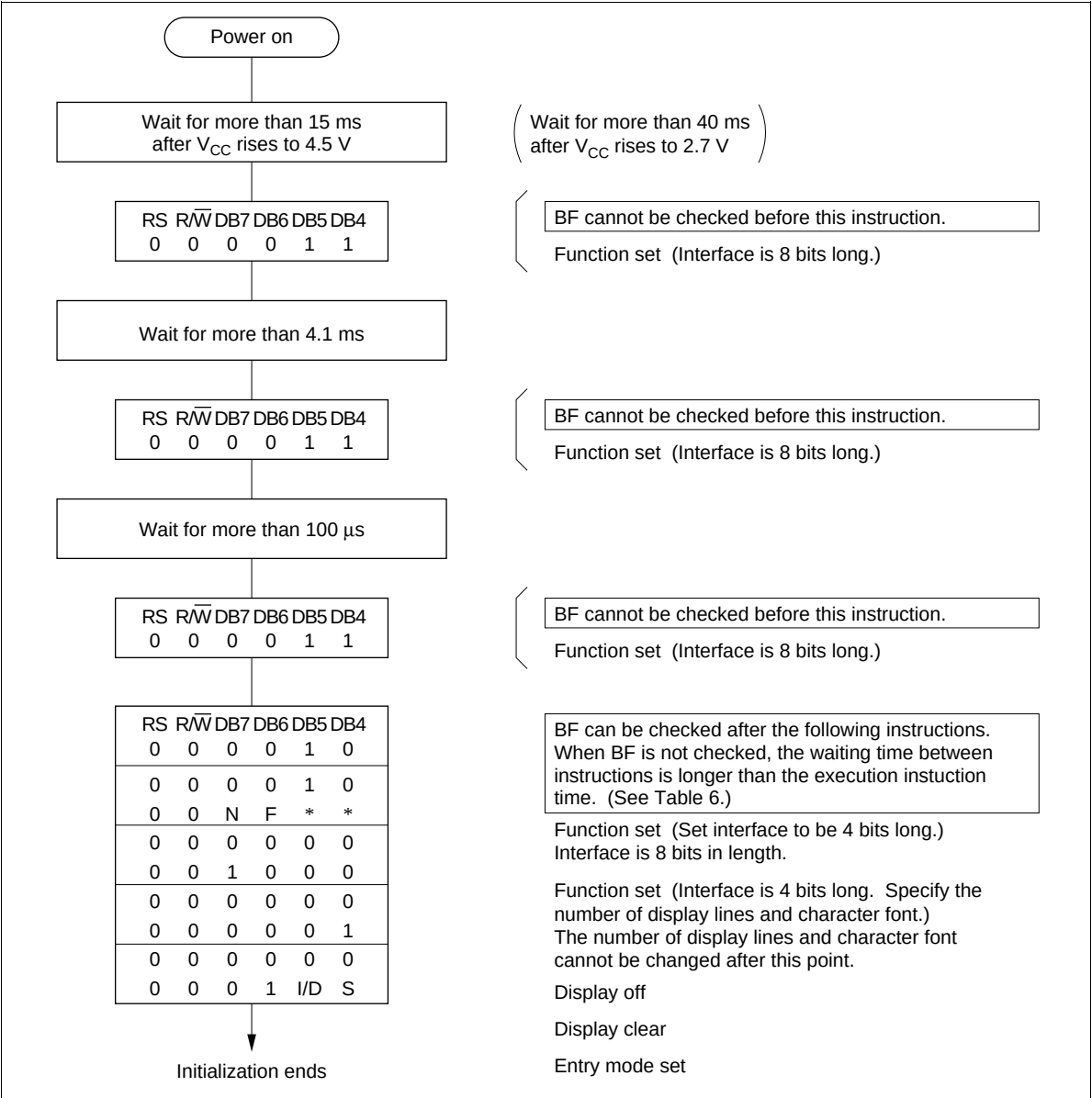


Figure 24 4-Bit Interface